

BANK MANAGEMENT SYSTEM

Professional Project README

Java Console Application + Responsive Web Interface

Project	Bank Management System
Java Files	Bank.java, MultiCustomerDemo.java
Web File	index.html
Account Types	Saving, Current
Maximum Accounts	50
Currency	Indian Rupee (₹)

1. Project Overview

The Bank Management System is a simple account-management project that supports creating customer accounts, viewing account details, depositing money, and withdrawing money. The Java implementation uses an array of Bank objects to manage multiple customer accounts, while the supplied HTML implementation presents the same core banking operations through a responsive dashboard.

The project is designed as a learning-oriented demonstration of object-oriented programming, console input handling, account operations, and a responsive web interface.

2. Features

- Open a new Saving or Current account.
- Generate a unique-style account number using the prefix **hdfc** followed by four digits.
- Set an opening balance of ₹10,000 for Saving accounts.
- Set an opening balance of ₹20,000 for Current accounts.
- Display account number, customer name, account type, and balance.
- Deposit money into an existing account.
- Withdraw money when sufficient funds are available.
- Search customer accounts by name or account number in the web interface.
- Responsive dashboard with account summary cards and transaction modals.

3. Technology Stack

Component	Technology
Core application	Java
Input handling	java.util.Scanner
Account number generation	java.util.Random
Web interface	HTML5, CSS3, JavaScript
UI framework	Bootstrap 5.3.3
Icons	Bootstrap Icons 1.11.3

4. Project Structure

Recommended GitHub repository structure:

```
Bank-Management-System/
■■■ Bank.java
■■■ MultiCustomerDemo.java
■■■ index.html
■■■ README.md
```

The Java classes separate account-level operations from the main multi-customer menu. The web interface is contained in index.html.

5. Java Application Workflow

Step	Operation
1	Start MultiCustomerDemo.
2	Choose Open Account.
3	Enter customer name and account type.
4	A generated account number and opening balance are assigned.
5	Use Account Enquiry to view account details.
6	Use Deposit or Withdraw with the account number.
7	Continue using the menu or exit.

6. How to Run the Java Version

1. Install a Java Development Kit (JDK) and make sure java and javac are available in the terminal.
2. Place **Bank.java** and **MultiCustomerDemo.java** in the same folder.
3. Open that folder in VS Code or Command Prompt.
4. Compile the source files:

```
javac Bank.java MultiCustomerDemo.java
```

5. Run the main program:

```
java MultiCustomerDemo
```

6. Follow the numbered menu shown in the console.

7. Main Menu

The Java program provides these operations:

Choice	Function
1	Open Account
2	Show Details / Account Enquiry
3	Deposit Money
4	Withdraw Money
5	Exit

8. Account Rules

The supplied Java implementation applies the following rules:

- Saving account → opening balance ■10,000.
- Current account → opening balance ■20,000.
- Maximum number of accounts in MultiCustomerDemo → 50.
- Withdrawal is rejected when the requested amount is greater than the available balance.
- An account number is generated in the form **hdfc####**.

9. Web Interface

The supplied index.html provides a responsive banking dashboard. It includes a navigation header, hero section, account summary cards, quick actions, customer-account table, search, and modal forms for account creation and transactions.

The dashboard displays Total Accounts, Total Balance, and Account Types. Users can open accounts, deposit money, withdraw money, view account details, and search accounts.

To run the web version, open **index.html** in a modern web browser. Because the supplied implementation is client-side, account data is stored in the browser session's JavaScript array and is not connected to a persistent database.

10. Web Application Flow

Action	Result
Open Account	Creates an account with customer name, type, generated number, and opening balance.
Deposit	Adds the entered amount to the selected account.
Withdraw	Subtracts the amount when sufficient balance is available.
View Details	Shows account number, customer name, account type, and available balance.
Search	Filters accounts by customer name or account number.

11. Sample Console Interaction

```
===== BANK MANAGEMENT SYSTEM =====
1. Open Account
2. Show Details (Account Enquiry)
3. Deposit Money
4. Withdraw Money
5. Exit
Enter Your Choice:
```

12. Learning Outcomes

This project demonstrates Java classes and objects, arrays of objects, methods, conditional logic, loops, switch-case menus, Scanner input, random account-number generation, and basic transaction logic. The web version additionally demonstrates responsive UI design, Bootstrap components, JavaScript DOM manipulation, form validation, search/filtering, and client-side state management.

13. Limitations of the Supplied Version

The supplied files do not show a database, authentication system, server-side API, or persistent storage. The Java implementation keeps up to 50 Bank objects in memory during execution, while the web implementation keeps accounts in a JavaScript array. Therefore, data persistence after the application/session ends is not part of the supplied implementation.

14. Future Enhancements

Possible extensions for a larger project include:

- Add a database such as MySQL or PostgreSQL for persistent account records.
- Add customer login and administrator authentication.
- Add transaction history and downloadable statements.
- Add backend REST APIs connecting the web interface to the Java or another server-side implementation.
- Add account deletion, transfer, interest calculation, and transaction search.
- Add stronger validation and server-side security for a production-oriented implementation.

15. GitHub README.md Version

For GitHub, use a Markdown file named **README.md** as the main repository documentation. The PDF version can be uploaded alongside it as **Bank_Management_System_README.pdf** for a printable project document.

16. Author

Chethla Sai Teja

B.Tech – Computer Science & Engineering (AI)

Project documentation prepared from the supplied Bank.java, MultiCustomerDemo.java, and index.html files.